
Quantum Computation and Communications (EE798V): Problem Set 6

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Q. 1: Prove the following elementary properties.

- (a) If $A, B \in \mathbb{C}^{d \times d}$, then $\text{Tr}(AB) = \text{Tr}(BA)$.
- (b) If $A, B \in \mathbb{C}^{d \times d}$ and $\alpha, \beta \in \mathbb{C}$, then $\text{Tr}(\alpha A + \beta B) = \alpha \text{Tr}(A) + \beta \text{Tr}(B)$.
- (c) If $A \in \mathcal{L}(\mathcal{H})$ and $\{|\psi_i\rangle\}_{i=1}^d$ is an orthonormal basis (ONB) of $\mathcal{H}$, then the following relation holds.

$$\text{Tr}(A) = \sum_{i=1}^d \langle \psi_i | A | \psi_i \rangle$$

- (d) If $A \in \mathbb{C}^{d_1 \times d_1}$ and $B \in \mathbb{C}^{d_2 \times d_2}$, then the following relation holds.

$$\text{Tr}(A \otimes B) = \text{Tr}(A) \text{Tr}(B)$$

- (e) If $\{|\psi_i\rangle\}_{i=1}^l, \{|\phi_j\rangle\}_{j=1}^m$ are ONBs of the Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$ respectively, then establish that $\{|\psi_i\rangle \otimes |\phi_j\rangle\}_{(i,j) \in \{1, \dots, l\} \times \{1, \dots, m\}}$ is an ONB of $\mathcal{H}_1 \otimes \mathcal{H}_2$.
- (f) Let $\mathcal{H}_1, \mathcal{H}_2$ be Hilbert spaces of dimension d . Use 1(e) to establish that any $|\xi\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$ can be expressed as follows.

$$|\xi\rangle = \sum_{i=1}^d |\chi_i\rangle \otimes |\phi_i\rangle$$

where $|\chi_i\rangle$'s are some elements of $\mathcal{H}_1$, and $\{|\phi_i\rangle\}_{i=1}^d$ is an ONB of $\mathcal{H}_2$.

- (g) If $\{|\phi_i\rangle\}_{i=1}^d$ is an ONB of $\mathcal{H}$, then prove that $\{|\phi_i\rangle \langle \phi_j|\}_{i,j \in \{1, \dots, d\}}$ is an ONB of $\mathcal{L}(\mathcal{H})$. In particular, establish that for any $A \in \mathcal{L}(\mathcal{H})$, the following relation holds

$$A = \sum_{i=1}^d \sum_{j=1}^d \alpha_{ij} |\phi_i\rangle \langle \phi_j|$$

where α_{ij} 's are some complex numbers. Obtain α_{ij} in terms of $|\phi_i\rangle, |\phi_j\rangle$, and A .

- (h) Let $\mathcal{H}_1, \mathcal{H}_2$ be Hilbert spaces of dimensions d_1 and d_2 respectively, and $C \in \mathcal{L}(\mathcal{H}_1 \otimes \mathcal{H}_2)$ denote an arbitrary linear operator acting on the Hilbert space $\mathcal{H}_1 \otimes \mathcal{H}_2$. Use 1(e) and 1(g) to demonstrate that

$$C = \sum_{i=1}^{d_1} \sum_{j=1}^{d_2} A_{ij} \otimes B_{ij}$$

where A_{ij} 's are some linear operators acting on the Hilbert space $\mathcal{H}_1$. Also, $B_{ij} = |\phi_i\rangle \langle \phi_j|$ where $\{|\phi_i\rangle\}_{i=1}^{d_2}$ is an ONB of $\mathcal{H}_2$.

Q. 2: Let $C_1, C_2 \in \mathcal{L}(\mathcal{H}_1 \otimes \mathcal{H}_2)$ where $\mathcal{H}_1, \mathcal{H}_2$ are Hilbert spaces associated with quantum systems S and E , respectively. Use 1(h) to establish the following results about the partial trace.

- (a) $\text{Tr}_E(\alpha C_1 + \beta C_2) = \alpha \text{Tr}_E(C_1) + \beta \text{Tr}_E(C_2)$ where $\alpha, \beta \in \mathbb{C}$.
- (b) If $A \in \mathcal{L}(\mathcal{H}_1)$, and $C \in \mathcal{L}(\mathcal{H}_1 \otimes \mathcal{H}_2)$, then $\text{Tr}_{SE}((A \otimes I_E)C) = \text{Tr}_S(A \text{Tr}_E(C))$. As a special case of this result, show that $\text{Tr}_{SE}(C) = \text{Tr}_S(\text{Tr}_E(C))$.
- (c) If $|\psi_1\rangle, |\psi_2\rangle \in \mathcal{H}_1$ and $\{|\phi_k\rangle\}_{k=1}^{d_2}$ is an ONB of $\mathcal{H}_2$, then exhibit that the following relation holds. You might have to use 1(c) to establish the result.

$$\langle \psi_1 | \text{Tr}_E(C_1) | \psi_2 \rangle = \sum_{k=1}^{d_2} \langle \psi_1 \otimes \phi_k | C_1 | \psi_2 \otimes \phi_k \rangle$$

- (d) Use 2(c) to prove that, if C_1 is a positive operator, then so is $\text{Tr}_E(C_1)$.

Let $\mathcal{H}_1, \mathcal{H}_2$ be Hilbert spaces of dimension 2, and the operator $C \in \mathcal{L}(\mathcal{H}_1 \otimes \mathcal{H}_2)$ is given as follows.

$$C = \begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.1 \\ 0.2 & 0.5 & -0.1 & 0 \\ -0.3 & 0.4 & 0.1 & 0.6 \\ 0.9 & 0.3 & 0.1 & 0.3 \end{bmatrix}$$

(e) Show that C can be decomposed as follows.

$$C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 0.1 & 0.2 \\ 0.2 & 0.5 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 0.3 & 0.1 \\ -0.1 & 0 \end{bmatrix} \\ + \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \otimes \begin{bmatrix} -0.3 & 0.4 \\ 0.9 & 0.3 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 0.1 & 0.6 \\ 0.1 & 0.3 \end{bmatrix}$$

(f) Using the above decomposition, compute $\text{Tr}_E(C)$ and $\text{Tr}_S(C)$.

(g) Verify that $\text{Tr}_S[\text{Tr}_E(C)] = \text{Tr}_E[\text{Tr}_S(C)] = \text{Tr}_{SE}(C)$.

(h) Show that C can alternatively be decomposed as follows.

$$C = \begin{bmatrix} 0.1 & 0.3 \\ -0.3 & 0.1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0.2 & 0.1 \\ 0.4 & 0.6 \end{bmatrix} \otimes \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \\ + \begin{bmatrix} 0.2 & -0.1 \\ 0.9 & 0.1 \end{bmatrix} \otimes \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 0.5 & 0 \\ 0.3 & 0.3 \end{bmatrix} \otimes \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

(i) Using the above decomposition, compute $\text{Tr}_E(C)$ and $\text{Tr}_S(C)$. Compare these results with 2(f).

Q. 3: Consider a composite qubit system composed of qubits S and E . Let ρ_{SE} be the density operator of the composite system. The symbol ρ_S denotes the reduced operator $\rho_S = \text{Tr}_E[\rho_{SE}]$.

(a) Using 2(b), show that $\text{Tr}_S[\rho_S] = 1$.

(b) Using 2(d), show that ρ_S is a positive operator.

(c) Combining 3(a) and 3(b), conclude that ρ_S is a valid density operator.

(d) Let $A \in \mathcal{L}(\mathcal{H}_S)$ be a Hermitian operator where the Hilbert space $\mathcal{H}_S$ describes the system S . Assume that the set $\{P_a\}_{a \in \sigma(A)}$ is the collection of all eigen-projections of A . Using 2(b), show the following.

$$\text{Tr}_S(\rho_S P_a) = \text{Tr}_{SE}((P_a \otimes I_E)\rho_{SE}), \quad a \in \sigma(A)$$

Use the above result to conclude that the reduced density operator ρ_S describes the system S .

(e) Let the density operator of the composite system is given as follows.

$$\rho_{SE} = \frac{1}{8} \begin{pmatrix} 1 & 1 & 1 & -1 \\ 1 & 5 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{pmatrix}$$

1. Verify that ρ_{SE} is a valid density operator, i.e., it is a positive operator with unit trace.
2. Compute the reduced density operators, ρ_S, ρ_E of the systems S and E , respectively.
3. Check whether the systems S and E are independent, i.e., whether ρ_{SE} equals $\rho_S \otimes \rho_E$.

Q. 4: Consider a composite system containing qubits S and E . It is known that the qubit S is described by the following density operator.

$$\rho_S = \begin{pmatrix} 0.5 & -0.25 \\ -0.25 & 0.5 \end{pmatrix} \quad (1)$$

(a) Check if ρ_S is a pure state operator.

(b) Obtain an eigen decomposition of ρ_S . What are the corresponding eigen projection operators?

(c) Obtain a pure state operator ρ_{SE} of the composite system such that the operator associated with the system S is given by (1). [Hint: Apply the purification protocol.]

Assume that a physical quantity is measured of the qubit E , given by the Hermitian operator, A stated below. For the following questions, assume that the density operator, ρ_{SE} of the composite system is what is calculated in 4(c).

$$A = \begin{pmatrix} 3 & \sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix}$$

(d) What are the outcome values and their respective probabilities?

(e) Compute the expectation and variance of A .

Q. 5: Consider a measurement M on a qubit system with state $|\psi\rangle$. The measurement M is such that a physical quantity described by the Hermitian operator σ_x is measured with 0.5 probability and another physical quantity described by another Hermitian operator σ_z is measured with 0.5 probability. Here σ_x and σ_z are standard Pauli matrices.

- Obtain the eigen-decompositions of σ_x and σ_z .
- Based on 5(a), what are the outcome values of the measurement M ?
- Show that the probability of a certain outcome a is given by $\langle\psi|\mathcal{E}_a\psi\rangle$ for some operator $\mathcal{E}_a$. Express these operators in the matrix form for each possible outcome.
- Show that the operators obtained in 5(c) are positive (i.e., Hermitian with non-negative eigenvalues).
- Show that the operators computed in 5(c) obey the completeness relation.
- Show that the operators obtained in 5(c) are not projection operators.
- Conclude that the measurement M cannot be described a Hermitian operator.
- Use 5(d) and 5(e) to prove that the set $\{\langle\psi|\mathcal{E}_a\psi\rangle\}_a$ indeed describes a valid probability distribution.

Q. 6: Let the density operator of a quantum system be ρ .

- If each constituent state of ρ is transformed by an unitary operator U , show that the resulting density operator is $U\rho U^\dagger$.
- Show that $U\rho U^\dagger$ is positive.
- Show that $\text{Tr}(U\rho U^\dagger) = 1$.
- Assume that the state ρ is transformed according to the unitary operator U_k with probability p_k where $p_k \geq 0, \forall k \in \{1, \dots, K\}$, and $\sum_k p_k = 1$. Show that the resulting density operator is given as:

$$\rho_f = \sum_{k=1}^K V_k \rho V_k^\dagger \quad (2)$$

where V_k 's are some operators such that

$$\sum_{k=1}^K V_k V_k^\dagger = I \quad (3)$$

- Show that ρ_f is a positive operator.
- Show that $\text{Tr}(\rho_f) = 1$.
A mapping of the density operators given by (2) and (3) is called a Kraus representation. Moreover, the operators $\{V_k\}_{k=1}^K$ are called Kraus operators.